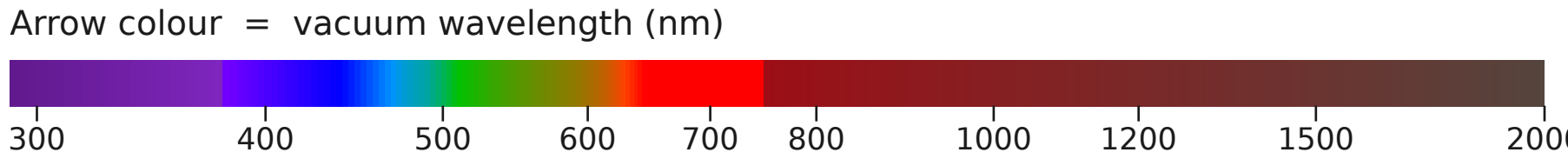


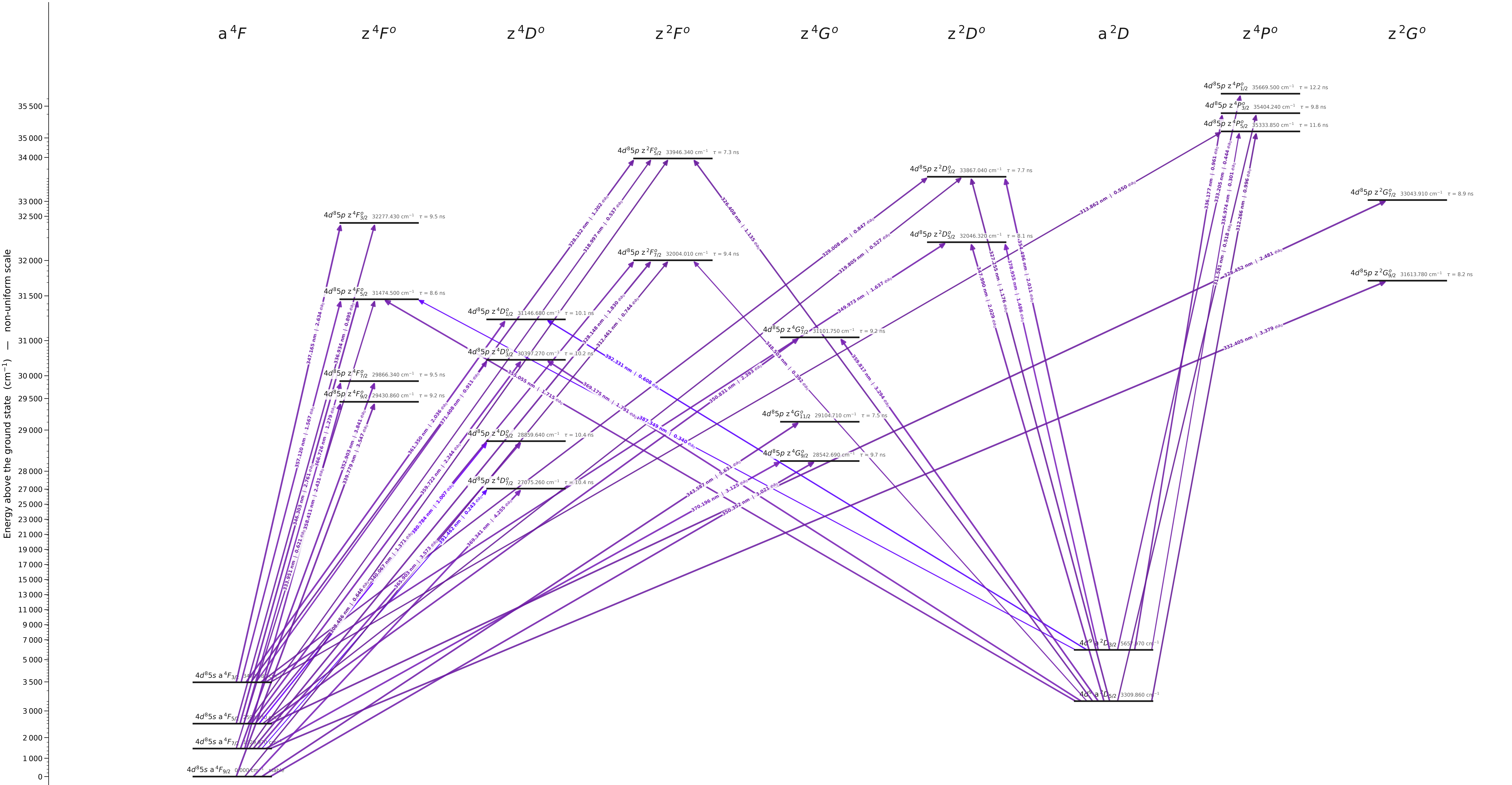
Rhodium

Arrow label: vacuum wavelength | reduced dipole matrix element $[U\|er\|J']$ in ea_0
Matrix elements are measured or NIST values; * = high-accuracy theory, $\approx$ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).



Rh (Rh I). Strongest classified transitions below $2\text{ }\mu\text{m}$ (NIST): 26 levels, 48 lines.

Rh $^+$ ionisation limit $60160.10\text{ cm}^{-1} = 7.45890\text{ eV}$ (166.223 nm)



Strongest transitions

Transition	λ vac (nm)	λ air (nm)	ν (THz)	$[U\ er\ J']$ (ea_0)	A (s $^{-1}$)	$\Gamma/2\pi$ up. (MHz)	branch (%)	Use
$4d^85s\ a^4F_{5/2} - 4d^85p\ z^2G^{\circ}_{7/2}$	328.4517	328.3571	912.7445	2.481	4.400e+07	17.88	39.2	
$4d^85s\ a^4F_{7/2} - 4d^85p\ z^2G^{\circ}_{9/2}$	332.4047	332.3091	901.8899	3.379	6.300e+07	19.41	51.7	
$4d^85s\ a^4F_{9/2} - 4d^85p\ z^4F^{\circ}_{9/2}$	339.7794	339.6819	882.3150	3.547	6.500e+07	17.3	59.8	
$4d^85s\ a^4F_{9/2} - 4d^85p\ z^4G^{\circ}_{11/2}$	343.5870	343.4885	872.5373	5.631	1.320e+08	21.22	99	
$4d^85s\ a^4F_{5/2} - 4d^85p\ z^4F^{\circ}_{9/2}$	346.3027	346.2036	865.6948	2.761	6.200e+07	18.51	53.3	
$4d^85s\ a^4F_{3/2} - 4d^85p\ z^4F^{\circ}_{9/2}$	347.1650	347.0656	863.5447	2.634	8.400e+07	16.75	79.8	
$4d^85s\ a^4F_{9/2} - 4d^85p\ z^4G^{\circ}_{9/2}$	350.3524	350.2522	855.6883	3.021	4.300e+07	16.41	41.7	
$4d^85s\ a^4F_{5/2} - 4d^85p\ z^4G^{\circ}_{7/2}$	350.8314	350.7311	854.5200	2.393	3.360e+07	17.3	30.9	
$4d^85s\ a^4F_{7/2} - 4d^85p\ z^4F^{\circ}_{7/2}$	352.9034	352.8025	849.5030	3.841	8.500e+07	16.75	80.8	
$4d^9\ a^2D_{3/2} - 4d^85p\ z^2D^{\circ}_{3/2}$	354.4959	354.3946	845.6866	2.011	4.600e+07	20.67	35.4	
$4d^85s\ a^4F_{5/2} - 4d^85p\ z^4D^{\circ}_{3/2}$	359.7221	359.6194	833.4003	2.244	5.480e+07	15.6	55.9	
$4d^9\ a^2D_{5/2} - 4d^85p\ z^4G^{\circ}_{7/2}$	359.8172	359.7146	833.1799	3.294	5.900e+07	17.3	54.3	
$4d^85s\ a^4F_{3/2} - 4d^85p\ z^4D^{\circ}_{1/2}$	361.3500	361.2470	829.6457	2.036	8.900e+07	15.76	89.9	
$4d^85s\ a^4F_{7/2} - 4d^85p\ z^4D^{\circ}_{5/2}$	365.9027	365.7985	819.3229	3.573	8.800e+07	15.3	91.5	
$4d^85s\ a^4F_{9/2} - 4d^85p\ z^4D^{\circ}_{7/2}$	369.3409	369.2358	811.6959	4.255	9.100e+07	15.3	94.6	
$4d^85s\ a^4F_{7/2} - 4d^85p\ z^4G^{\circ}_{9/2}$	370.1960	370.0906	809.8210	3.125	3.900e+07	16.41	37.8	

λ , ν from NIST level energies unless a measured frequency for this isotope is available. A: measured value or NIST.
 $\Gamma/2\pi$: natural linewidth of the upper level, $1/(2\pi\tau)$.